

Alpilla 600 MW Combined Cycle Power Plant

Piping Design and Layout Criteria

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1. Purpose and scope

1.1 This document defines the design and layout criteria for all piping of the power plant inside the battery limits: process, steam, water, fuel, chemical and utility piping, above ground and underground, from the equipment nozzles and the tie-in points of the interface register to the terminal points.

1.2 Piping classes (materials, ratings, branch tables, corrosion allowances) are defined in the Piping Material Classes ALP-EPC-00000-PI-SPC-0001; welding, heat treatment and NDE in ALP-EPC-00000-PI-SPC-0003; the stress analysis criteria and the critical line list in ALP-EPC-00000-PI-RPT-0001; insulation in ALP-EPC-00000-ME-SPC-0003; painting in ALP-EPC-00000-ME-SPC-0002. This document shall not repeat their values.

2. Codes and standards

2.1 Power piping (steam, feedwater, condensate, boiler blowdown and drains, auxiliary steam, cooling water inside the power block, compressed air, fuel gas and LDO within the plant boundary) shall be designed to ASME B31.1.

2.2 Process piping in chemical, water treatment and ammonia service shall be designed to ASME B31.3 (normal fluid service; category M only when the toxicity assessment requires it). Gas pipework of the gas receiving and metering station from the terminal point to the fiscal metering and pressure control station outlet shall be designed to ASME B31.8 and EN 12186 as applicable; downstream fuel gas piping shall be designed to ASME B31.1.

2.3 Piping components shall comply with ASME B16.5 (flanges up to NPS 24), B16.34 (valves), B16.25 (butt-welding ends) and B36.10M (pipe dimensions); larger flanges to ASME B16.47 series A. Pressure equipment directive category shall be shown on the class sheets for piping in the European Union scope.

2.4 Seawater piping in GRP shall be designed to a recognised standard (EN 1796 / ISO 14692 or AWWA M45 for buried pipes) as defined in ALP-EPC-00000-PI-SPC-0007.

2.5 Where the codes give different rules, the more stringent shall apply; deviations shall be approved by IEPC. Interface piping to the IEC power island shall follow the interface data of the tie-in register and the OEM design data.

3. Tie-in points and interfaces

3.1 All tie-in points at the interface with the IEC scope and with the Owner (gas, LDO, water, CW, steam, electrical) are defined in the tie-in register with size, rating, material, pressure and temperature. Piping design shall start from these values and shall not change them without a change in the register.

Point	Service	Status
IF-01	Natural gas to GT	agreed
IF-02	LDO to GT	agreed
IF-03	Demin water for water injection	agreed
IF-04	Compressor wash water	agreed
IF-05	HP steam	agreed
IF-06	HP steam	agreed
IF-07	Cold reheat	agreed
IF-08	Cold reheat	agreed
IF-09	Hot reheat	agreed
IF-10	Hot reheat	agreed
IF-11	LP steam	agreed
IF-12	LP steam	agreed

Point	Service	Status
IF-13	HP / IP / LP bypass valves	agreed
IF-14	Bypass steam to condenser	agreed
IF-15	Condensate from hotwell	agreed
IF-16	Condensate to CPH	agreed
IF-17	BFP suction	agreed
IF-18	HP feedwater	agreed
IF-19	IP feedwater	agreed
IF-20	FGH water return	agreed
IF-21	HRSG blowdown	agreed
IF-22	HRSG drains and vents	agreed
IF-23	Chemical dosing	agreed
IF-24	Sampling	agreed
IF-25	Aqueous ammonia	agreed
IF-26	Flue gas to stack	agreed
IF-27	Circulating water	agreed
IF-28	Closed cooling water	agreed
IF-29	Instrument and service air	agreed
IF-30	Nitrogen	agreed
IF-31	Hydrogen / CO2	agreed
IF-32	Auxiliary steam	agreed
IF-33	Fire protection	agreed
IF-34	GT GSU HV	agreed
IF-35	ST GSU HV	agreed
IF-36	UAT supply	agreed
IF-37	MV supplies	agreed
IF-38	LV supplies	agreed
IF-39	DC and UPS	agreed
IF-40	Control system interface	agreed
IF-41	Protection interface	agreed
IF-42	Earthing	agreed
IF-43	Foundations	agreed
IF-44	Condenser tube cleaning balls	agreed
IF-45	HP attemperation spray water	agreed
IF-46	Reheat attemperation spray water	agreed
TP-A1	Temporary construction area (5 ha)	agreed
TP-A2	Site (plot) handover	agreed
TP-CW1	Seawater intake head	agreed
TP-CW2	Outfall diffuser	agreed
TP-D1	Storm water discharge	agreed
TP-E1	Grid connection 380 kV	agreed
TP-E2	Construction power	agreed
TP-G1	Natural gas supply	agreed
TP-L1	LDO supply	agreed
TP-M1	Temporary barge landing	agreed
TP-R1	Road access	agreed
TP-S1	Sanitary wastewater	agreed
TP-T1	Telecommunication	agreed
TP-T2	Emission data link to the environmental authority	agreed
TP-T3	TSO SCADA and revenue metering link	agreed
TP-W1	Potable water	agreed

4. Design pressure and design temperature

4.1 The design pressure of a line shall be the highest coincident of the following: maximum operating pressure plus 10 % (minimum 1.0 bar), the set pressure of the relief valve protecting the system, the shut-off head of the pump feeding the system, and the static head plus pressure at the tie-in; for steam and feedwater systems, the design pressure of the HRSG-connected piping is the drum or header design pressure of the equipment register.

4.2 The design temperature of a line shall be the maximum operating temperature plus 15 K for steam and water systems above 150 degC, plus 10 K for other systems, and minimum design metal temperature (MDMT) per the lowest ambient temperature of the site (DP-CASE-0009) or the lowest process temperature (Joule-Thomson cooling in gas pressure reduction), whichever is lower.

4.3 Design pressure and design temperature of lines connected to equipment shall be not lower than the equipment design data (design_pressure, design_temperature in the equipment register) up to and including the first isolation valve, and shall agree with the piping class; lines shall not be rated lower than their relief device set pressure.

4.4 Vacuum service shall be identified on the line list; piping and supports under vacuum (condenser, steam turbine exhaust) shall be designed for full vacuum with 1.5 bar external pressure and shall be checked for buckling.

4.5 Line list entries (design and operating pressure and temperature, test pressure, class, insulation type, heat tracing) shall be extracted from the piping database; design values in the line list shall not be typed but shall be taken from the registers and the process data.

4.6 Hydrostatic test pressure shall be 1.5 x design pressure (adjusted for temperature per ASME B31.1); where hydrotest of steam lines can overload the supports, a temporary support or a pneumatic test with approved procedure shall be used. Seawater GRP piping shall be tested per manufacturer's instructions.

5. Line sizing

5.1 Lines shall be sized on the maximum flow of the heat and mass balance cases with the margins of the equipment sizing basis; the velocity and pressure drop limits below apply to maximum flow, and the smaller of the velocity and the pressure-drop criterion governs.

Service	Velocity / pressure drop limit	Remarks
HP steam, cold reheat, hot reheat (superheated)	30 to 60 m/s; pressure drop 0.3 bar/100 m for HP steam, 0.2 bar/100 m for reheat	Total pressure drop of cold reheat and hot reheat as per the HMB limit [DEC-EPCE-0014]
LP steam and auxiliary steam	25 to 50 m/s (saturated 20 to 35 m/s); 0.2 bar/100 m	Noise limit by check for control valve outlet piping
Steam to condenser (bypass)	Up to 80 m/s at the condenser nozzle with noise / vibration check	Sparger design per the condenser vendor
Boiler feedwater, suction	0.8 to 1.5 m/s	NPSH margin governs
Boiler feedwater, discharge	2.0 to 4.0 m/s; 0.5 bar/100 m	
Condensate, suction	0.5 to 1.2 m/s	Hotwell pump NPSH governs
Condensate, discharge	1.5 to 2.5 m/s	
Circulating seawater (GRP / lined CS)	1.0 to 2.5 m/s; 3.0 m/s maximum in fittings	Above 3.0 m/s erosion, below 0.8 m/s fouling; velocity at sizing checked against the condenser tube velocity of DP-OWNR-0023 [DP-OWNR-0023, DEC-EPCE-0002]
Closed cooling water, demin water	1.0 to 2.5 m/s (demin water up to 2.0 m/s in stainless steel)	
Fire water	Up to 3.0 m/s in mains; 5 m/s in laterals at maximum demand	Per NFPA 20 / NFPA 24; design flow DP-EPCE-0006 [DP-EPCE-0006, REF-CONS-0070]
Fuel gas	15 to 20 m/s (maximum 20 m/s downstream of pressure control); 0.2 bar/100 m	Noise and vibration check for regulator outlet piping
LDO	Suction 0.5 to 1.0 m/s; discharge 1.0 to 2.0 m/s	Viscosity at the minimum temperature
Aqueous ammonia, hypochlorite and chemicals	0.5 to 1.5 m/s	Minimum 25 mm for clogging where dosing lines are of lower flow, otherwise 20 mm
Compressed air and nitrogen	10 to 15 m/s; 0.1 bar/100 m	
Drain and waste (gravity)	0.6 to 2.5 m/s, filling ratio maximum 70 %; minimum slope 1:100	
Two-phase, wet steam and flash lines	Maximum 25 m/s with erosion check; velocity per flow regime	FAC-sensitive lines are in the critical line list [ALP-EPC-00000-PI-RPT-0001]

6. Layout: pipe racks, elevations and clearances

6.1 Piping shall be arranged in pipe racks along the main routes with piperack tiers by service: steam and hot lines on the upper tier, process and utility water on the intermediate tier, cables and instrument on the separate trays; each rack shall have at least 20 % spare space and a 25 % spare load capacity for future lines.

6.2 Rack elevations and clearances: minimum headroom under pipe racks over roads is 6.0 m (heavy transport routes 7.5 m), over the access ways and walkways 2.5 m, over platforms 2.2 m; clear vertical distance between pipe racks and cable trays is 0.3 m; clearance around insulated lines shall include insulation and cladding.

6.3 Space between parallel pipes shall allow flanges to be staggered with a minimum clear distance of 50 mm between the outside of the insulation of adjacent lines and 100 mm from steel; pipes shall be routed with bends and expansion loops as required by the stress analysis, rather than expansion joints (expansion joints are used only where approved by IEPC).

6.4 Piping shall be routed in the 3D model, without interference with the equipment maintenance envelopes, crane paths, doors and escape routes; escape routes shall keep the widths of HS-DBR-0001 section 9 and be free of pipes below 2.2 m.

6.5 Access and maintenance: valves, instruments, sampling points and flanges shall be accessible from grade, platforms or ladders; all operating and maintenance points above 2.2 m shall be reachable from a platform; extended stems or chains are not accepted, except for valves in remote locations approved by IEPC.

6.6 Handwheel and operating elevations: manual valves shall have the handwheel centre between 0.8 m and 1.5 m above the operating level (maximum 1.8 m), with the stem not projecting into the walkway; manual valves above 1.8 m shall be operated from a platform; motor-operated valves shall have clear access to the actuator and handwheel.

6.7 Pipework shall be laid out to avoid pockets; drains shall be provided at each low point and vents at each high point of the pipework; hydrotest vents and drains shall be DN 20 minimum (DN 25 for lines above DN 300), with double isolation and blind flange for HP and high-temperature lines.

6.8 Piping connected to rotating equipment shall be arranged to avoid excess nozzle loads, with a removable spool at each pump suction; suction piping shall have an eccentric reducer flat side up, and straight run of 5 diameters at the pump suction.

6.9 Underground piping shall be minimised and shall be laid at least 1.0 m below the finished ground (0.8 m with protection) and with a marker tape; crossings of roads and rails shall use sleeves or concrete-encased pipes. Trenches shall be provided with the bedding sand, warning tape and accessible chambers for valves.

7. Supports, expansion and stress analysis

7.1 Pipe stress analysis shall follow ALP-EPC-00000-PI-RPT-0001, which contains the critical line list and the load cases (sustained, thermal expansion, occasional loads wind and seismic, water hammer, safety valve discharge). The wind speed and seismic load shall be those of the site data.

7.2 Seismic loads for piping and supports shall be based on the site seismic parameters (PGA of DP-CASE-0002, site class ZD) per TBDY 2018; lines are analysed with the response spectrum or an equivalent static coefficient for the piping in the critical line list.

7.3 Supports shall be of standard types (hangers, guides, stops, anchors, spring supports) with hot-dip galvanised carbon steel outdoors (EN ISO 1461) and C5 coating; supports for stainless steel and GRP shall be isolated from the pipe with a suitable pad. Spring hangers shall have a load variation of maximum 25 %.

7.4 Loads on equipment nozzles shall be within the allowable loads of the vendor (or ISO 13707 / API 610, API 617, NEMA SM23) and shall be verified in the pipe stress analysis; pump suction nozzle loads at 2 x API 610 table values where approved by the pump vendor.

7.5 The design of steam piping shall consider the number of thermal cycles: 250 starts per year over the design life, with cold, warm and hot start cycles of the operating regime. Fatigue and creep shall be assessed for P91 and P22 lines in accordance with the code.

8. Special systems

8.1 Steam piping shall be arranged with drainage on all low points and dead legs with steam traps or manual drains, with continuous slope 1:250 in the direction of flow (minimum 1:100 for drain lines); warming and drain systems shall be arranged for the start-up regime of the plant (hot, warm and cold starts).

8.2 Fuel gas piping shall be all-welded with flanges only at equipment and valves; the piping shall be laid above ground on sleepers or racks with clearances to ignition sources, with double block and bleed at the GT connection; the pressure drop shall be compatible with the load limiter curve of the gas turbine at the tie-in point IF-01.

8.3 LDO piping shall have leak detection at the flanges, and shall be arranged with trace heating as required for the pour point of the LDO; the piping shall be protected from spilled fuel and shall have slope towards the recovery sumps.

8.4 Seawater piping, underground, shall be GRP, in accordance with ALP-EPC-00000-PI-SPC-0007 and ALP-EPC-00000-PI-MRQ-0004, with concrete thrust blocks or restraints for bends, buried with sand bedding and protection against traffic load; steel fittings (flanges, valves) shall be lined or of super duplex per ALP-EPC-00000-ME-RPT-0002.

8.5 Aqueous ammonia, hypochlorite and other chemicals shall be in double-contained pipe or in pipes routed in trays with banded collection; pipe routes shall be separated from the electrical equipment, and shall be marked with the service, direction of flow and hazard sign.

8.6 Safety valve discharge piping shall be designed for the reaction force and shall discharge to a safe location; steam relief valves discharge through silencers to atmosphere at least 3 m above any platform within 15 m; discharge shall not be directed at other equipment.

9. Insulation, tracing and identification

9.1 Insulation of piping shall be in accordance with ALP-EPC-00000-ME-SPC-0003 for personnel protection above 60 degC surface temperature at reachable heights (up to 2.2 m above the platform and 1.0 m sideways), for energy conservation and for freeze protection where required by the site minimum ambient temperature (DP-CASE-0007).

9.2 Piping identification (service name, flow arrow, colour band) shall follow ALP-EPC-00000-ME-SPC-0002 and the KKS numbering of the project; lines shall be tagged in the piping design database with the KKS number of the system.

10. Quality, models and deliverables

10.1 Piping design shall be produced in the 3D model with the model review stages (30 %, 60 %, 90 %) and clash detection; pipes of DN 50 and larger are modelled, small-bore piping (below DN 50) may be routed on site with approved standards.

10.2 Isometric drawings, pipe support drawings and stress analysis reports shall be generated per system from the piping model and shall be given to the construction by the work packages defined in the AWP plan; the piping input from Plant 3D or E3D shall be imported into the database in the PCF format.

10.3 Material take-off of pipes, valves and fittings shall be generated from the model, with a wastage of 5 % (10 % for small bore) plus the site spare; the valve list (ALP-EPC-00000-PI-LST-0001) shall be consistent with the piping and instrumentation diagrams.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
TBDY 2018	Turkish Building Earthquake Code (Türkiye Bina Deprem Yönetmeliği)
PED 2014/68/EU	Pressure Equipment Directive, transposed as the Pressure Equipment Regulation (Basıncılı Ekipmanlar Yönetmeliği)
ASME B31.1	Power Piping
ASME B31.3	Process Piping
ASME B31.8	Gas Transmission and Distribution Piping Systems
EN 12186	Gas infrastructure - Gas pressure regulating stations for transmission and distribution
ASME B16.5	Pipe Flanges and Flanged Fittings NPS 1/2 through NPS 24
ASME B16.25	Butt welding Ends
ASME B16.34	Valves - Flanged, Threaded, and Welding End
NFPA 20	Standard for the Installation of Stationary Pumps for Fire Protection
EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles
ASME B36.10M	Welded and Seamless Wrought Steel Pipe